```
#include<stdio.h>

int graph[10][10],visited[10],queue[10];
int n,front=-1,rear=-1;

void bfs(int start)
{
   int i;
   front=rear=-1;
   queue[++rear]=start;
   visited[start]=1;

   while(front!=rear)
   {
      start=queue[++front];
      printf("%d ",start);

      for(i=0;i<n;i++)
      {
         if(graph[start][i]==1 && visited[i]==0)
         {
            queue[++rear]=i;
            visited[i]=1;
         }
      }
   }
}

int main()
{
   int i,j,start;

   printf("Enter number of vertices: ");
   scanf("%d",&n);

   printf("Enter adjacency matrix:\n");

   for(i=0;i<n;i++)
   {
      for(j=0;j<n;j++)
      {
         scanf("%d",&graph[i][j]);
      }
   }
```

```
    printf("Enter starting vertex: ");
    scanf("%d",&start);

    bfs(start);

    return 0;
}

}
```

OUTPUT

main.c

Clear Run ▶

```c
#include<stdio.h>

int graph[10][10],visited[10],queue[10];
int n,front=-1,rear=-1;

void bfs(int start)
{
    int i;
    front=rear=-1;
    queue[++rear]=start;
    visited[start]=1;

    while(front!=rear)
    {
        start=queue[++front];
        printf("%d ",start);

        for(i=0;i<n;i++)
        {
            if(graph[start][i]==1 && visited[i]==0)
            {
                queue[++rear]=i;
                visited[i]=1;
            }
        }
    }
```

```c
}

int main()
{
    int i,j,start;

    printf("Enter number of vertices: ");
    scanf("%d",&n);

    printf("Enter adjacency matrix:\n");

    for(i=0;i<n;i++)
    {
        for(j=0;j<n;j++)
        {
            scanf("%d",&graph[i][j]);
        }
    }

    printf("Enter starting vertex: ");
    scanf("%d",&start);

    bfs(start);

    return 0;
}
```

```
Enter num
Enter adj
1 2 3 4 5
89908776
445668899
```